Module 7: Introduction



Syllabus (https://northeastern.instructure.com/courses/257815/assignments/syllabus)
Modules (https://northeastern.instructure.com/courses/257815/modules)
Resources (https://northeastern.instructure.com/courses/257815/pages/course-resource-page-2)
Assignments (https://northeastern.instructure.com/courses/257815/assignme

AI Foundations Course - Midterm Examination - Requires Respondus LockDown Browser + Webcam, Quiz

Module Overview

This module introduces students to the field of computer vision, covering both classical techniques and modern deep learning approaches. Students will learn the fundamentals of image processing, feature detection and description, object detection and recognition, image segmentation, and video analysis. The module covers popular architectures, including YOLO, R-CNN, and U-Net, as well as applications in face recognition, autonomous vehicles, and medical imaging.

Learning Objectives

By the end of this module, you should be able to:

- Get familiar with image processing techniques for enhancement and feature extraction
- Understand principles of object detection and classification systems
- Learn about image segmentation models for various applications
- Understand computer vision pipelines for real-world problems
- Learn about 3D vision and video analysis techniques

Task List

- Participate in the weekly discussion board
- Develop and submit the midterm project
- Additional reading:
  - **Foundational Texts**
    - Hartley, R., & Zisserman, A. (2003). *Multiple View Geometry in Computer Vision* (2nd ed.). Cambridge University Press.

- Orts-Escolano, S., Riemenschnitter, M., Volkhardt, M., & Stühmer, L. (2016). *Computer Vision for Augmented and Virtual Reality*. KIT Scientific Publishing. https://doi.org/10.5445/KSP/1000057053 (https://www.google.com/search?q=https://doi.org/10.5445/KSP/1000057053)
- Goodfellow, I., Bengio, Y., & Courville, A. (2016).[1] *Deep Learning*. MIT Press.
- Schroff, F., Kalenichenko, D., & Philbin, J. (2015). FaceNet: A Unified Embedding for Face Recognition and Clustering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 815–823). https://doi.org/10.1109/CVPR.2015.7298682 (https://doi.org/10.1109/CVPR.2015.7298682)
- Taigman, Y., Yang, M., Ranzato, M. A., & Wolf, L. (2014). DeepFace: Closing the Gap to Human-Level Performance in Face Verification. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 1701–1708). https://doi.org/10.1109/CVPR.2014.220 (https://doi.org/10.1109/CVPR.2014.220)
- Krizhevsky, A., Sutskever, I., & Hinton, G. E. (2012). ImageNet Classification with Deep Convolutional Neural Networks. In F. Pereira, C. J. C. Burges, L. Bottou, & K. Q. Weinberger (Eds.), *Advances in Neural Information Processing Systems 25 (NIPS 2012)* (pp. 1097–1105).
- He, K., Gkioxari, G., Dollár, P., & Girshick, R. (2017). Mask R-CNN. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)* (pp. 2961-2969). https://doi.org/10.1109/ICCV.2017.322 (https://doi.org/10.1109/ICCV.2017.322)
- Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. In N. Navab, J. Hornegger, P. F. Fischer, & A. M. Ladd (Eds.), *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015* (pp. 234–241). Springer, Cham. https://doi.org/10.1007/978-3-319-24574-4_28 (https://doi.org/10.1007/978-3-319-24574-4_28)
- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.
- Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). You Only Look Once: Unified, Real-Time Object Detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 779-788).

Applications

- Rizzoli, A. (2021, July 25). *27+ Most Popular Computer Vision Applications*. V7 Labs. https://www.v7labs.com/blog/computer-vision-applications?utm_source=chatgpt.com (https://www.v7labs.com/blog/computer-vision-applications?utm_source=chatgpt.com)
- Janai, J., Günther, M., Schönberger, F., Gehrig, M., & Schindler, K. (2020). Computer Vision for Autonomous Driving: Problems, Datasets and State of the Art. *Foundations and Trends® in Computer Graphics and Vision*, *12*(1-2), 1–170. https://doi.org/10.1561/0600000078 (https://www.google.com/search?q=https://doi.org/10.1561/0600000078)
- MarketsandMarkets. (2025, May 16). *Computer Vision in Healthcare Market worth US$14.39 billion by 2030 with 24.3% CAGR*. PR Newswire. https://www.marketsandmarkets.com/PressReleases/computer-vision-healthcare.asp (https://www.marketsandmarkets.com/PressReleases/computer-vision-healthcare.asp)

Recent Research

- Szeliski, R. (2022). *Computer Vision: Algorithms and Applications* (2nd ed.). Springer.
- Zhang, W., Li, X., Liu, X., Lu, S., & Tang, H. (2025). Facing challenges: A survey of object tracking. *Digital Signal Processing*, *161*. https://www.sciencedirect.com/science/article/pii/S1051200425001046 (https://www.sciencedirect.com/science/article/pii/S1051200425001046)
- Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., & Wallach, H. (2020). *Datasheets for datasets*. Communications of the ACM, *63*(12), 86-92.
- Smith. (2024, October 24). *Transforming Sight into Insight: Master Computer Vision!* Medium. https://medium.com/@asbmerin/transforming-sight-into-insight-master-computer-vision-66c0e7fc64cd (https://medium.com/@asbmerin/transforming-sight-into-insight-master-computer-vision-66c0e7fc64cd)
- Torralba, A., Isola, P., & Freeman, W. T. (2024). *Foundations of Computer Vision*. MIT Press.

Online Resources

- Gonzalez, R. C., & Woods, R. E. (2018). *Digital Image Processing* (4th ed.). Pearson.
- Chen, J., & Ran, X. (2019). Deep Learning With Edge Computing: A Review. *Proceedings of the IEEE*, *107*(8), 1655–1674. https://doi.org/10.1109/JPROC.2019.2921977 (https://doi.org/10.1109/JPROC.2019.2921977)
- Nixon, M. S., & Aguado, A. S. (2019). *Feature Extraction in Computer Vision* (3rd ed.). Academic Press.
- One Hundred Year Study on Artificial Intelligence (AI100). (n.d.). *A Short History of AI*. Stanford University. Retrieved July 19, 2025, from https://ai100.stanford.edu/2016-report/appendix-i-short-history-ai (https://ai100.stanford.edu/2016-report/appendix-i-short-history-ai)
- Murphy-Chutorian, E., & Trivedi, M. M. (2009). *Real-time Computer Vision Systems*. Springer.